

Miao Xu

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EDUCATION

Lund University Master in Machine Learning, Systems and Control (Sweden)(Full Scholarship) • Courses: Learning-based Control (5), Monte Carlo and Empirical Methods for Stochastic Inference (5), Computer Vision (5), Introduction to ANN and Deep Learning (4), Modelling and Learning from Data (5).	2025 – Present
ITMO University & Hangzhou Dianzi University (dual degree) Bachelor in Control in Technical Systems with Honours (2 %) (Russia) & Bachelor in Automation (China) • GPA: 5.0/5.0 (Russia), 4.69/5.0 (China) (Rank: 2/93) • Courses: Higher Mathematics (98), Theoretical Mechanics (100), Identification Theory (100), Automatic Control Theory (98), Adaptive and Robust control (100), Nonlinear Control Systems (100), Optimal Control (100).	2019 – 2023

PUBLICATIONS

[1] COVER: cross-vehicle transition framework for quadrotor control in air-ground cooperation. [Paper] [Media]
Ren, Q., Xu, M., Zhang, M., Chen, N., Lai, M., Xu, C., Gao, F. and Cao, Y., 2025. *Autonomous Robots*, 49(3), p.23.

RESEARCH EXPERIENCE

FAST Lab, Zhejiang University Advisor: Yanjun Cao Research Assistant	Zhejiang, China 2023 – 2025
• Air-Ground Cooperative (AGC) Framework for Inter-Vehicle Transfer Tasks – Developed a non-inertial multi-frame AGC framework and differential model for GPS-free UAV transfer between moving UGVs without high-rate replanning. – Designed a phase-adaptive Model Predictive Controller (MPC) with a novel position reference scheme. – Paper [1] has been published in <i>Autonomous Robots</i>.	
• Communication-Free Robust UAV Tracking and Landing on Dynamic Platforms – Developed a robocentric tracking model using only inter-robot relative visual measurements. – Built an L_1-adaptive MPC framework that estimates unmeasured platform motion, removing the need for motion-state data from the ground platform; real-flight validation is ongoing.	

PROJECT EXPERIENCE

Safe Reinforcement Learning for Dynamic Bimanual Manipulation CS Project, Lund University Advisor: Davide Tateo	Lund, Sweden 2026
• Built an Isaac Sim digital twin of a dual-arm KUKA iiwa setup for bimanual peg-in-hole manipulation. • Formulated the task as a constrained RL problem and benchmarked SAC against Lagrangian SAC. • Source code released at https://github.com/XuMiaoMiaoMiao/Safe_RL_Peg_in_hole .	
Detect empty drums on the exterior walls of buildings using a quadrotor Intelligent sound detection project Advisor: Chao Xu , Yanjun Cao	Zhejiang, China 2023 – 2024
• Designed an innovative quadrotor capable of flying closely along walls, moving smoothly, and tapping surfaces for sound signal collection, offering an effective solution for detecting hollow drums in building exteriors.	
Research on Control Algorithm of Quadcopter Undergraduate Thesis Advisor: Ke Yin, Sergei Shavetov	Zhejiang, China 2022 – 2023
• Designed a cascade active disturbance rejection controller for robust quadrotor tracking under disturbances. • Validated robustness via numerical simulations and effectiveness through hardware-in-the-loop tests. • Thesis: Research on Control Algorithm of Quadcopter (Outstanding Undergraduate Thesis)	

AWARDS

Excellent Graduate of Zhejiang Province (4 %)	2023
Outstanding Paper Award , "SHENZHEN CUP" Math Modeling Challenge (4 %)	2021
National Second Prize for China Aeromodelling Design Challenge (15 %)	2020 & 2021
Scholarship of Zhejiang Province (5 %)	2020 & 2021 & 2022
The First Prize Scholarship of Hangzhou Dianzi University (5 %)	2020 & 2021 & 2022

SKILLS

Programming: C++ (4+ years), C (6+ years), Python (5+ years), MATLAB (5+ years), PyTorch, MushroomRL.
Software & Hardware Development: Linux, ROS, Gazebo, Isaac Sim, Git, Docker, SolidWorks, AutoCAD.